

RFID Attendance Logger

Power, Capacity and Cost

Current draw for each operating mode and event, when the MCU and the reader are awake, the reasoning behind each clock frequency, the expected number of card reads and the battery life on a 1000 mAh, 3.7 V Li-ion cell, how the STM32L432KB's 128 kB of flash is shared between the program and the attendance log, and what a finished unit costs to make at 1 to 10 000 units, in US dollars and Sri Lankan rupees.

Hardware	Battery	Date	Status
STM32L432KB (128 kB) + ST25R3916	1000 mAh, 3.7 V nominal (3.7 Wh)	08 October 2026	Currents measured 8 October 2026

BATTERY LIFE 17 months 4 lectures/day, 100 students, incl. self-discharge (12 pessimistic)	IDLE SCANNING 108 μ A Reader wake-up mode is 91 % of it	SWITCHED OFF 1.8 μ A Standby, measured; 57 years on the device alone	PER CARD TAP 2.8 μ Ah 10.3 mC; 6,858 records fit in flash
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1 Summary

The logger spends nearly all of its life in one of two low-power states. Switched off, it sits in Standby and draws **1.8 μ A** (measured). Switched on and waiting for cards, the MCU sleeps in Stop 2 and wakes only for a 10 ms heartbeat flash every 5 s, while the ST25R3916 checks the antenna itself every 100 ms. That idle state draws about **108 μ A**, and 91 % of it is the reader's wake-up mode. The large currents (202 mA for the RF field, 82 mA for the motor) flow only for milliseconds per tap, so a whole 100-student lecture costs about **0.36 mAh** (1.34 mWh).

At four lectures a day, the device uses about **33 mAh a month**. With the cell's self-discharge (about 20 mAh a month) that gives roughly **17 months per charge**, or 27 months from the device's draw alone.

The program and the log share the STM32L432KB's 128 kB flash: 72 kB for the program (60.6 kB at -Og) and 56 kB for the log, **6,858 records** or about 64 lectures of 100 students. The log fills long before the battery empties (in about 16 teaching days at four lectures a day), so the PC sync, which clears the log, sets the routine, not the charger.

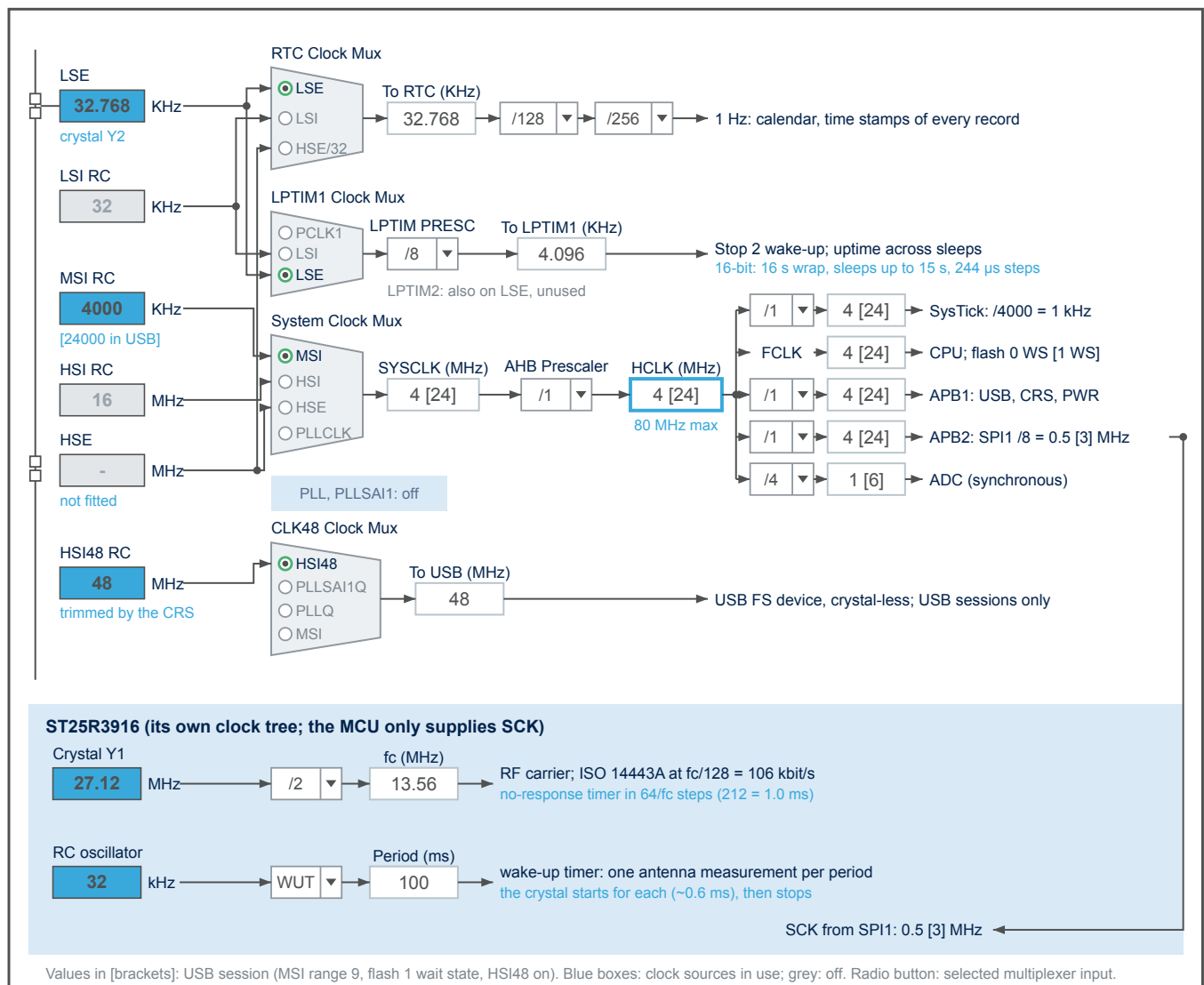
A finished unit costs about **US\$25 (LKR 8,382)** to make at 1 000 units and US\$18 (LKR 6,042) at 10 000, landed in Sri Lanka and tested (section 11).

Confidence. Every current in this report except the 24/80 MHz run currents, the ADC and the flash was measured on the board (8 October 2026). Steady loads were read with a multimeter in series with the cell, using a power-test build that holds each load steady; the pulses (a wake-up measurement, a card tap, a false wake-up) were captured with a Tektronix TBS1052C across a 1 ohm shunt and integrated (section 3). The two methods disagree on the reader's wake-up mode: the scope's 9.9 μC per measurement gives 99 μA , the multimeter, which cannot follow 94 mA spikes on its μA range, read 117 - 204 μA . The typical column uses the scope, the pessimistic one the multimeter's 185 μA and 4 false wake-ups a minute. The RF field current is for the untuned antenna.

2 Clock frequency selection

Each clock was picked for the job it does. The scanning clock matters most, because it runs on every wake-up.

Clock tree



Drawn with STM32CubeMX's own clock-configuration palette and font, but with the values the firmware programs at run time (bsp_clock.c, bsp_time.c, bsp_nfc.c), which override the .ioc. Values in brackets apply during a USB session (MSI range 9, flash 1 wait state, HSI48 on). The LSE branch keeps running in Stop 2 and Standby; everything fed from SYSCLK stops in Stop 2. SPI1's divider is the peripheral's own baud-rate prescaler, not an RCC one.

Clock domain	Selection	Setting	Why
Scanning (SYSCLK/HCLK)	MSI range 6, 4 MHz	Range 1, 0 wait states	Reset clock and the Stop 2 wake-up clock (STOPWUCK = MSI): no PLL to rellock on each wake-up. The work is time-bound (5 ms field guard, 106 kbit/s frames, ADC sampling), so a faster clock only raises the current for the same window.
USB session	MSI range 9, 24 MHz	Range 1, 1 wait state	USB FS needs HCLK above 14.2 MHz (RM0394). 24 MHz is the first MSI step with margin for rendering CSV sectors on demand. Powered from VBUS, so it costs the battery nothing.
USB kernel clock	HSI48 trimmed by CRS	Locks to the host's 1 kHz SOF	Meets the USB ± 0.25 % tolerance with no crystal and no PLL. Stopped outside USB sessions.
RTC and Stop 2 timer	LSE 32.768 kHz	RTC; LPTIM1 at LSE / 8 (16-bit)	Runs through Stop 2 for well under 1 μ A. LPTIM1 counts 4096 Hz, so its 16-bit count wraps every 16 s and one Stop 2 sleep can last up to 15 s (the loop asks for at most 5 s); resolution 244 μ s. A wake from the wrap alone goes straight back to Stop 2.
Reader SPI	4 MHz / 8 = 500 kHz	Mode 1, ST25R3916 max 10 MHz	Well inside the chip's limit. Register traffic during a read adds an estimated 2-3 ms of field-on time; a /2 prescaler (2 MHz) would cut it to under 1 ms.
ADC	HCLK / 4 = 1 MHz	640.5-cycle sample	0.64 ms per channel gives the 1.7 M Ω divider time to settle; the ADC is powered only for each sample.
Reader crystal	27.12 MHz	2 $\times$ 13.56 MHz carrier	Fixed by ISO 14443 (fc = 13.56 MHz). Runs only in Ready mode or for each wake-up measurement.
Reader wake-up timer	ST25R3916 32 kHz RC	100 ms period	Keeps the 27.12 MHz crystal off between measurements, so the reader idles at about 3 μ A between them. 100 ms is the slowest check a person will not notice; each check costs about 10 μ C, so the period sets most of the idle current.
SysTick	HCLK / 4000 = 1 kHz	Stopped in Stop 2	HAL's millisecond tick, the firmware's only time base while awake. LPTIM1 adds the time slept in Stop 2, so timers never see a gap.
Calendar	LSE / 128 / 256 = 1 Hz	RTC, backup domain	ST's lowest-power split of the prescalers. Keeps the date and time through Standby and resets for the attendance time stamps.
Clocks left off	PLL, PLLSAI1, HSI16, LSI	-	No load needs more than 24 MHz, so there is no PLL to start or lock (15-40 μ s, plus current). There is no HSE crystal. LSI would only be needed by a watchdog, which is not enabled. MSI is not locked to the LSE (MSI PLL mode off): nothing timed depends on it.

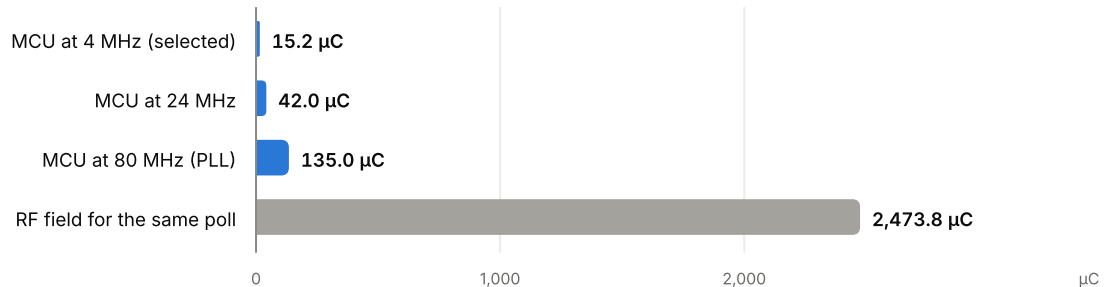
Why 4 MHz and not faster

The MCU's awake time is set by the radio and the protocol rather than by computation: a read waits out a 5 ms field guard, exchanges ISO 14443 frames at 106 kbit/s, and busy-waits on the reader's IRQ line. For a fixed window t , the charge is $Q = I_{run}(f) \times t$, so the lowest clock that meets the deadlines wins.

MCU charge during one 15 ms read window (4 MHz measured, 24/80 MHz datasheet):

4 MHz	=	1.01 mA × 15 ms	=	15.2 μC
24 MHz	=	2.8 mA × 15 ms	=	42.0 μC
80 MHz	=	9.0 mA × 15 ms	=	135.0 μC + PLL lock per wake-up
RF field	=	202 mA × 12 ms	=	2474 μC

Charge for one card read: MCU clock options against the RF field



At 4 MHz the MCU is 0.6 % of a read's charge. At 80 MHz it would be about 5 %, for no gain in speed.

For purely computational work (one pass of the main loop is roughly 2000 cycles), current per MHz is close to constant, so finishing faster and sleeping longer gains little: 0.5 ms at 1.0 mA (0.51 μC) at 4 MHz against 25 μs at 9 mA (0.23 μC) at 80 MHz. That small gain is lost to the PLL's start-up time and to the 4 flash wait states 80 MHz needs.

Why not slower

- **SPI with the field on.** The SPI clock is SYSCCLK / 8. At 2 MHz the reader's register traffic would take twice as long while the 202 mA field is on, costing more than the MCU saves.
- **ADC sampling.** The ADC clock is HCLK / 4. A slower clock stretches the 640.5-cycle sample and keeps the ADC powered longer.
- **Low-power run (at most 2 MHz) would barely help.** The MCU is awake about 0.02 % of the time in idle, so its run current adds only ~0.2 μA to the 108 μA idle figure.

Voltage scaling

The firmware keeps Range 1 at all clocks. Range 2 would cut run current by about 10 %, but that saves well under 0.1 μA in idle, while USB needs Range 1 anyway. Staying in Range 1 avoids switching the regulator on every USB attach. Stop 2 and Standby use the low-power regulator either way.

3 Measurements and assumptions

Battery-side currents, measured on 8 October 2026 with a multimeter in series with the cell. The power-test build (make power-test, Bsp/Src/bsp_power_test.c) runs no application: each press of the button holds the next load steady, so the meter reads it instead of a millisecond pulse. The 3.3 V rail comes from a linear LDO, so the battery supplies the same current the MCU and LEDs draw; the reader's VDD/VDD_TX and the motor sit straight on BAT+.

Measured MEASURED 8 OCTOBER 2026

Mode	State held	Current	Note
1	MCU Stop 2 + RTC + LPTIM1, reader power-down	3.3 μ A	Includes the divider, LDO and charger leakage
2	As 1, reader in wake-up mode (100 ms)	103 μ A	Meter read 117-204 μ A as it averaged the 100 ms measurements; 188 averaged
3	As 1, reader Ready (oscillator on, field off)	5.66 mA	Reader between polls and during patterns
4	As 1, RF field on	202 mA	Antenna not yet tuned
5	MCU Run, MSI 4 MHz, reader power-down	1.01 mA	DS11451 gives ~0.5 mA; peripheral clocks are on
6	MCU Sleep, MSI 4 MHz, SysTick every 1 ms	0.72 mA	The loop's mode in USB sessions
7	Green LED on (470 Ω from 3.3 V)	2.77 mA	
8	Red LED on	3.0 mA	
9	Vibration motor on (coin ERM on BAT+)	56.5 mA	
10	Standby (switched off)	1.8 μ A	4.8 μ A before the reader's SPI pins were pulled down in Standby

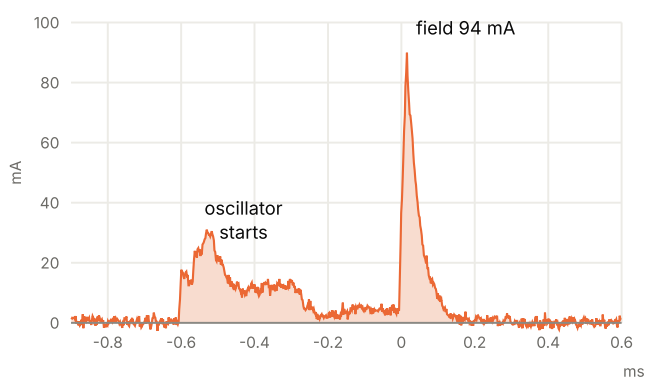
Scope captures MEASURED 8 OCTOBER 2026

Tektronix TBS1052C, probe and menu at 10X, across a 1 ohm shunt in the battery's negative lead, so 1 mV is 1 mA; USB and the debugger unplugged. Each capture's DC offset (up to 2 mA) is taken from a quiet stretch and subtracted, and the charge is the integral of the current. The files are in Waveforms/; Firmware/docs/waveforms.py does the arithmetic.

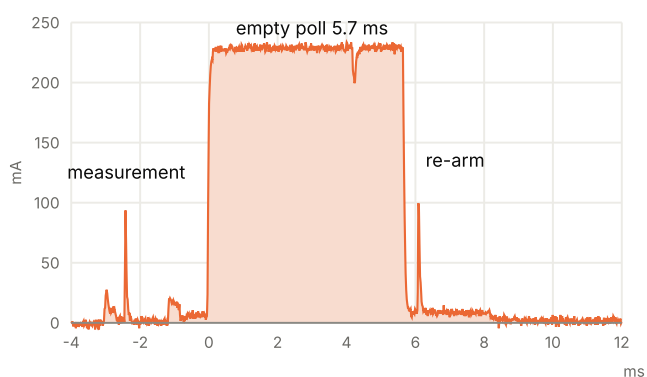
Event	Capture	Charge / current	Detail
One wake-up measurement	TEK00005	9.9 μ C	Oscillator start 13 mA $\times$ 0.4 ms, plateau 3.7 mA, field spike 94 mA peak for 41 μ s; two more in TEK00007: 5.9 and 8.8 μ C
Wake-up mode average		99 μ A	9.9 μ C $\times$ 10 per second, plus the reader's timer (a few μ A, below the scope's resolution)
Wake-up IRQ to field on	TEK00006	25.9 μ C	The measurement that saw the card, MCU wake, oscillator start on
Read: field on	TEK00006	2.47 mC	12.2 ms at 202 mA (guard, REQA, anticollision, SELECT)

Event	Capture	Charge / current	Detail
Buzz	TEK00006	4.60 mC	50 ms at 91 mA: motor 82 mA with start-up, Ready, LED
Ready + green LED	TEK00006	9.24 mA	After the buzz (multimeter: 5.66 + 2.77 mA)
Empty poll	TEK00007	1.30 mC	5.7 ms at 228 mA
Re-arm (amplitude measure)	TEK00007	14.5 μ C	
False wake-up, IRQ	TEK00007	1.33 mC	Next measurement 105 ms later, as expected to re-arm

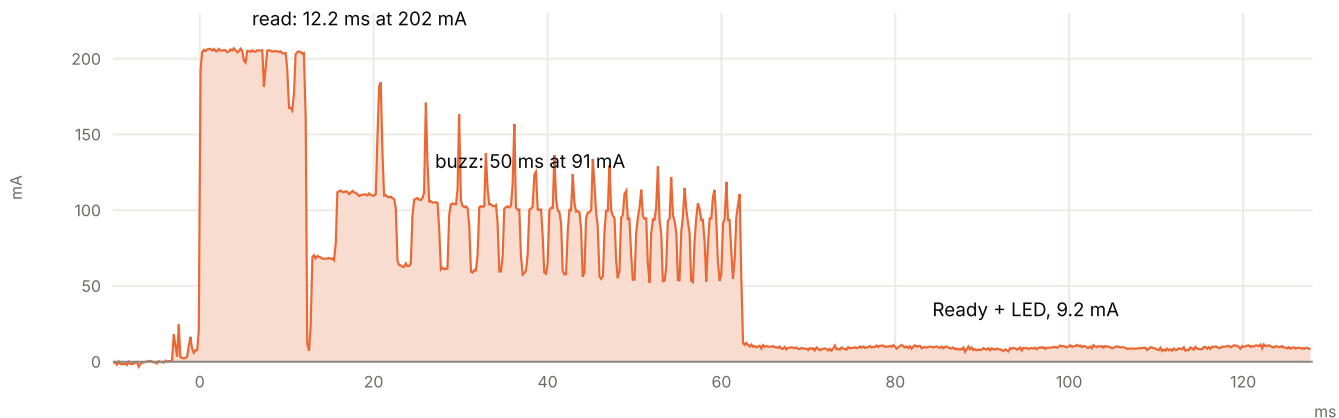
One wake-up measurement (TEK00005)



A false wake-up (TEK00007)



A card tap (TEK00006)



Battery current with the scope's offset removed. The capture ends at 128 ms; the LED stays on, with the reader in Ready mode, to the end of the 250 ms pattern, and one empty poll then re-arms wake-up mode (as in the false wake-up). The buzz measured 50 ms against the 60 ms set: the pattern's clock starts at the beginning of the loop pass that read the card, before the ~6 ms read. The motor averages 82 mA over the buzz against 56.5 mA running, because it spends the buzz starting up.

Changes made after the first measurements

Change	Before	After
Debug clocks off in Stop 2 / Standby (DBGMCU_CR cleared at boot)	0.3 mA switched off	4.8 μ A
Reader SPI, LED and motor pins pulled down in Standby	4.8 μ A switched off	1.8 μ A
Heartbeat flash	30 ms every 4 s	10 ms every 5 s
Longest idle sleep (APP_SLEEP_MAX_MS); LPTIM1 / 8	1 s (1.35 wakes/s)	5 s (0.4 wakes/s)

Change	Before	After
Battery sample placed on the heartbeat's wake	own wake every 10 s	none extra
Empty polls before re-arming (APP_NFC_REMOVE_MISSES)	3	1
Accepted-card buzz (APP_FB_ACCEPT_VIB_MS)	90 ms	60 ms

Still estimated

Quantity	Typical	Pessimistic	Source / basis
STM32L432 Run, 24 MHz / 80 MHz	2.8 / 9.0 mA	-	DS11451 typ., Range 1 (not used while scanning)
ADC on, extra over Run	0.2 mA	-	DS11451
Flash program / erase	7.0 mA	-	DS11451; 22 ms per page erase
Button pull-ups (R8 10k + internal)	0.41 mA	-	Only while pressed
Field time per read / per empty poll	12.2 / 5.7 ms	14.1 / 6.6 ms	Scope (one capture each); pessimistic +15 %
Reader wake-up mode average	99 µA	185 µA	Scope (pulse charge × 10/s) against the multimeter
False wake-ups while scanning	1 / min	4 / min	Not measured yet; read dbg_nfc_false_wakes
Cell self-discharge	2 %/month	3 %/month	Typical Li-ion at room temperature
Usable capacity of 1000 mAh	900 mAh	850 mAh	Cut-off at 3.3 V, ageing margin

Measured on the board MEASURED 07 OCTOBER 2026

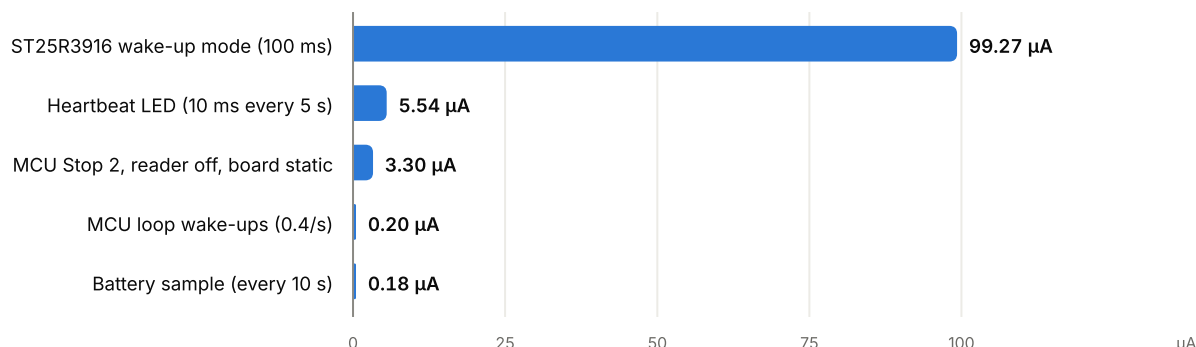
Quantity	Value	Note
Flash size (0x1FFF75E0)	128 kB	Fitted part is the STM32L432KB
Cell voltage (dbg_battery_mv)	4,152 mV	Three samples 10 s apart: 4152, 4149, 4154 mV
State of charge (dbg_battery_percent)	96 %	From the firmware's Li-ion voltage curve
ADC counts at PA7	1,876	PA7 = 3308 mV × 1876 / 4095 = 1515 mV
VDDA from VREFINT	3,308 mV	VREFINT_CAL 1652 / 1497 counts × 3000 mV
Divider check	4,153 mV	1515 mV × (4.7 M + 2.7 M) / 2.7 M

4 Operating modes

Mode	Typical	Pessimistic	What is running
Off (Standby)	1.8 µA	1.8 µA	MCU Standby + RTC, reader power-down, divider (measured)
Idle, scanning	108 µA	194 µA	Stop 2, reader wake-up mode, heartbeat
Polling mode (wake-up off)	18.9 mA	-	Reader Ready + field every 100 ms; for comparison
Card read (RF field on)	202 mA	-	12.2 ms per read, 228 mA in an empty poll (scope)
Vibration motor on	82 mA	-	Average over a 50 ms buzz (scope); 56.5 mA running steadily

Mode	Typical	Pessimistic	What is running
USB session	4 - 6 mA	-	From VBUS, not the battery (plus charge current)

Idle scanning current, 108 µA in total



The reader's own antenna measurements are 91 % of idle current. The MCU, including its wake-ups, is under 4 %.

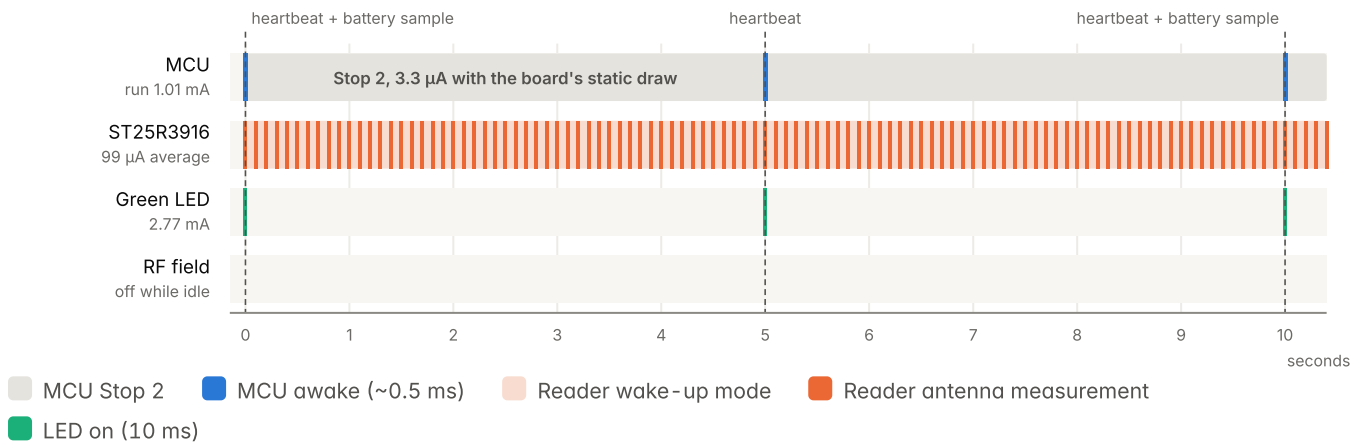
Wake-up mode	=	$9.93 \mu\text{C per measurement} \times 10 / \text{s (scope)}$	=	99.3 µA
Heartbeat	=	$2.77 \text{ mA} \times 10 \text{ ms} / 5000 \text{ ms}$	=	5.5 µA
Wake-ups	=	$0.4 / \text{s} \times 0.5 \text{ ms} \times 1.01 \text{ mA}$	=	0.20 µA
Sampling	=	$1.5 \text{ ms} \times (1.01 + 0.2) \text{ mA} / 10 \text{ s}$	=	0.18 µA
Off	=	measured, mode 10	=	1.80 µA
Polling mode	=	$5.66 \text{ mA Ready} + 228 \text{ mA} \times 5.7 \text{ ms} / 100 \text{ ms} + \text{MCU}$	=	18.9 mA

The reader's wake-up mode is still far cheaper than polling: switching it off would take idle current from 108 µA to about 19 mA, and battery life left switched on from 346 days to 2.0 days. Each measurement costs 9.9 µC (scope), ten times the ~1 µC first assumed: it starts the 27.12 MHz oscillator (5.0 µC) and drives the untuned antenna for about 41 µs at up to 94 mA (4.0 µC). A longer period (section 10) or a tuned antenna is the largest remaining saving.

5 When the MCU and the reader are awake

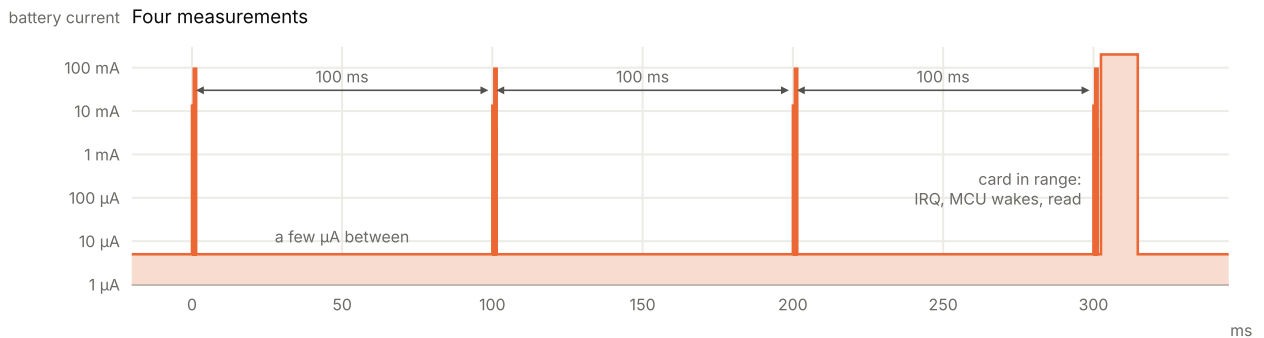
The two chips sleep independently. The MCU sleeps in Stop 2 and wakes on its own timer (LPTIM1) or on an interrupt: the button, VBUS, or the reader's IRQ line. The ST25R3916 runs its own wake-up timer and wakes the MCU only when the antenna changes. Timings are the firmware's; the currents beside each lane are the measured ones.

Idle scanning: ten seconds



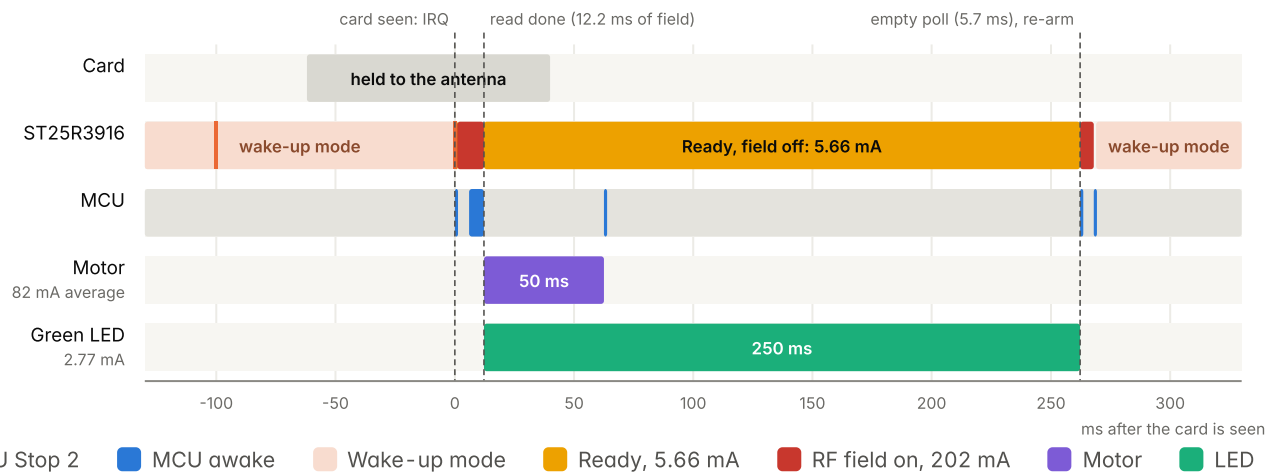
The MCU wakes twice every 5 s: when the heartbeat LED goes on and 10 ms later when it goes off. The battery is sampled on the first of those wakes every 10 s, so it adds no wake of its own. Meanwhile the reader measures the antenna 10 times a second, without the MCU. Nothing else is due, so the loop sleeps the full 5 s (APP_SLEEP_MAX_MS).

The reader's wake-up measurements, every 100 ms



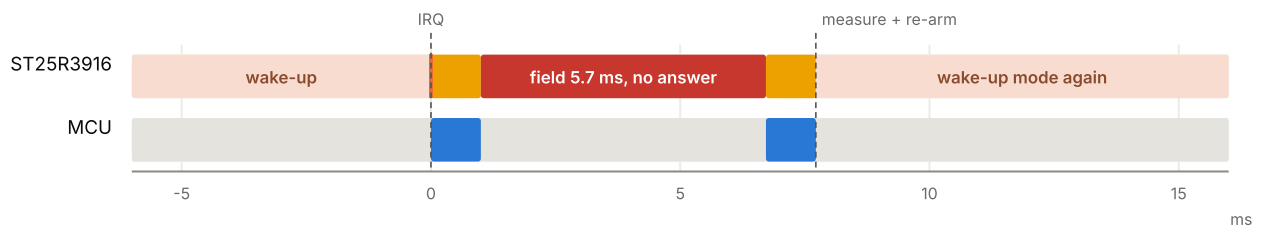
Log scale; each pulse is drawn from the measured one (section 3): the oscillator starts at about 13 mA for 0.4 ms, then the antenna is driven for about 41 μ s at up to 94 mA, 9.9 μ C in all. Ten times a second the ST25R3916 compares the amplitude with its reference while the MCU sleeps. When a card changes it past the window (the fourth measurement), the reader raises its IRQ and the MCU wakes to read the card (12.2 ms of field).

One accepted card tap



The card is noticed at the reader's next measurement, at most 100 ms after it arrives. The IRQ wakes the MCU, which turns the field on and sleeps through the 6 ms guard in Stop 2, then runs REQA, anticollision and SELECT (the field is on 12.2 ms in all, measured), logs the card and starts the pattern. The reader stays in Ready mode while the pattern plays, so the motor never runs with the field on. At the end of the pattern one empty poll (the card has gone) re-arms wake-up mode. Five seconds after the last tap the MCU wakes once more to write the RAM buffer to flash.

A false wake-up

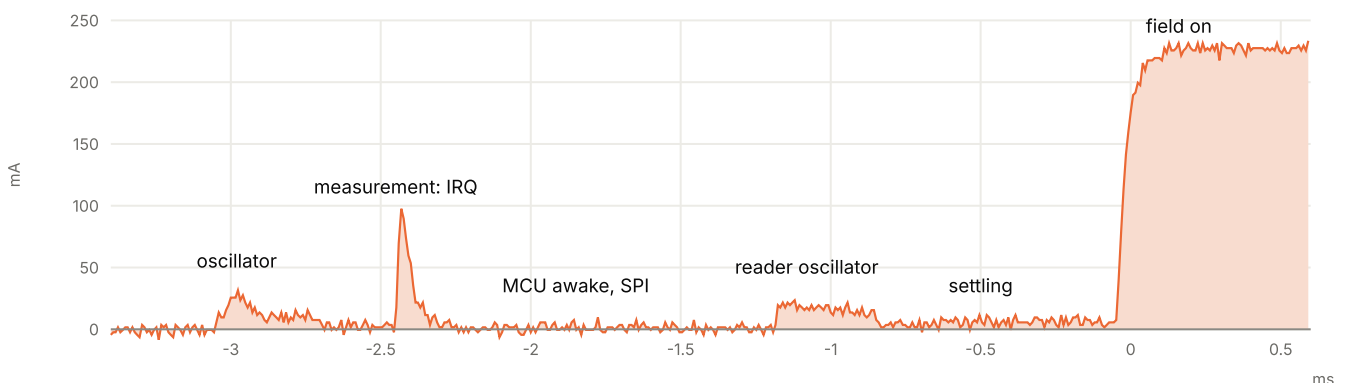


A measurement strays past the window with no card there (noise, a hand, drift). One empty poll, 5.7 ms of field at 228 mA (measured), and the reader is armed again with a fresh reference: 1.33 mC each, against about 5.5 mC with the three empty polls used before.

Wake-up, settling and sleep-entry times

What each transition costs in time. The datasheet figures bound the hardware; the measured ones are what the firmware does on this board, so the gaps between them are the firmware's own work (loop passes, SPI traffic at 500 kHz, waiting for the reader's crystal).

From a wake-up measurement to the field: a false wake-up, zoomed in (TEK00007)



The measurement that strays raises the IRQ. The MCU leaves Stop 2 in about 13 μ s, but takes 1.3 ms to read the reader's status and restart its oscillator over SPI at 500 kHz, and another 1.2 ms passes before the field comes on. A faster SPI clock (section 10) would shorten the first part.

Measured on the board	Time	Source / what happens
Wake-up measurement: oscillator start to field	605 μ s	TEK00005
Wake-up measurement: whole event	731 μ s	TEK00005
Measurement to the reader's oscillator restarting	1.28 / 1.27 ms	MCU leaves Stop 2 (~13 μ s), runs the loop, reads the reader over SPI at 500 kHz
Oscillator restart to field on	1.15 / 1.22 ms	Crystal settles (I _{osc}), next loop pass
Card seen to field on	2.43 / 2.49 ms	False wake-up / card tap
Field on for a read (guard, REQA..SELECT)	12.25 ms	TEK00006
Field on for an empty poll	5.69 ms	TEK00007; see the note below
Field off to re-arm measurement	392 μ s	TEK00007
Re-arm to the next wake-up measurement	98.8 ms	TEK00007: the 100 ms wake-up timer
Buzz (motor on)	50.1 ms	TEK00006; 60 ms set

STM32L432 (DS11451 Rev 4)	Typ	Max	Where it applies
Sleep to Run	6 CPU cycles	6 cycles	1.5 μ s at 4 MHz; USB sessions and short waits
Stop 2 to Run, MSI 4 MHz, from flash	12.06 μ s	13.16 μ s	Every idle wake-up (datasheet lists 4 MHz at Range 2)
Stop 2 to Run, MSI 48 MHz, Range 1	8.02 μ s	9.24 μ s	For comparison: a faster wake clock saves 4 μ s
Standby to Run, MSI 4 MHz	19.14 μ s	25.8 μ s	Power button; then reset and start-up code
Shutdown to Run	261.5 μ s	315.7 μ s	Not used: Shutdown would lose the RTC
Reset after power-on (BOR)	250 μ s	400 μ s	Battery connected
Regulator Range 2 to Range 1	20 μ s	40 μ s	Not used: the firmware stays in Range 1
MSI start-up, ranges 4-7	3 μ s	6 μ s	MSI restarts on every Stop 2 exit
MSI PLL-mode settling to 1 %	-	2.5 ms	Not used (MSI PLL mode off)
HSI48 start-up	2.5 μ s	6 μ s	USB attach; the CRS then trims it on each 1 ms SOF
PLL lock	15 μ s	40 μ s	Not used (PLL off)
LSE start-up	2 s	-	Crystal dependent; only when the battery is connected (firmware allows 5 s)
ADC regulator start-up	-	20 μ s	Every battery sample
ADC calibration	116 ADC cycles	-	116 μ s at 1 MHz, every sample
VREFINT buffer start	8 μ s	12 μ s	Every sample; then at least 4 μ s sampling
Flash: one 64-bit record	81.69 μ s	90.76 μ s	Each log record

STM32L432 (DS11451 Rev 4)	Typ	Max	Where it applies
Flash: page erase (2 kB)	22.02 ms	24.47 ms	Once per 254 records, and #CLEARLOG

ST25R3916 (DS12484 Rev 8)	Typ	Max	Note
Power -down to Ready: crystal start-up	crystal dependent	-	L _{osc} at 750 mVpp; firmware waits up to 50 ms
Wake-up mode: field per measurement	~20 µs	-	Measured: 41 µs above 40 mA
Measure amplitude command	-	25 µs	Each re-arm
Adjust regulators command	-	5 ms	Start-up and a supply-mode change
ISO 14443 guard, field on to first command	5 ms	-	Firmware: 5 ms + 1 tick (APP_NFC_FIELD_GUARD_MS)
No-response timer	$212 \times 64/f_c = 1.0 \text{ ms}$	-	Ends an empty poll (BSP_NFC_NRT_64FC)
Supply: power-down	2 µA	20 µA	Switched off
Supply: wake-up mode (logic + RC)	2.8 µA	20 µA	Between measurements; 3.6 µA in the datasheet's example
Supply: Ready	4.5 mA	7.5 mA	Measured 5.66 mA

Firmware	Time	What it covers
Enter Stop 2	<100 µs	Mask interrupts, re-check the queue and inputs, write the LPTIM1 compare (synchronises in ~3 LSE cycles, 92 µs), WFI
Shortest / longest Stop 2 sleep	4 ms / 15 s	Shorter waits use Sleep mode; the loop asks for at most 5 s (BSP_STOP2_MIN/MAX_MS, APP_SLEEP_MAX_MS)
Enter Standby	~1 s	Power-off pattern (700 ms), flush to flash (up to 24.5 ms with an erase), reader power-down, wait for the button release
Button debounce / hold / off / double press	30 ms / 2 s / 5 s / 600 ms	APP_BTN_*
VBUS debounce; USB enumeration timeout	50 ms; 5 s	No enumeration in 5 s means a charger
Battery divider settling	1.2 s	C3 through R7 R8 = 171 ms time constant (APP_BATT_SETTLE_MS)
Flush RAM records	5 s after the last tap	Or at 80 % full, on USB attach, before power-off
Inactivity switch-off	3 min	Suspended while VBUS is present

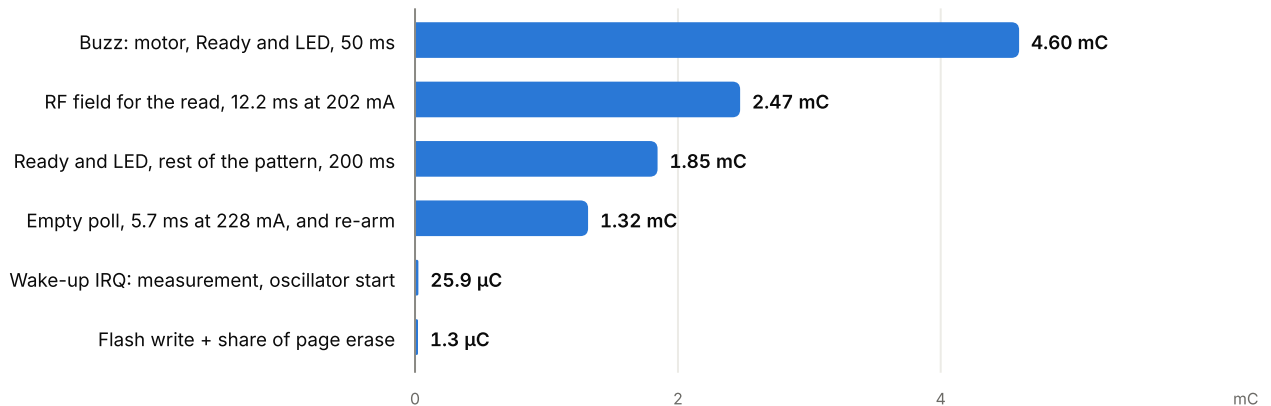
Open finding: the field guard looks short. An empty poll keeps the field on 5.7 ms in all, although the firmware waits 5 ms + 1 tick before REQA and the no-response timer then adds about 1 ms. The buzz is also shorter than set (50 ms against 60). Both suggest that the millisecond count runs ahead across short Stop 2 sleeps, or that the guard starts before the field does, so the real guard may be under the 5 ms ISO 14443 requires. Cards still read, but it should be confirmed with a GPIO marker on the scope. Timing the guard with the reader's own NFC field-on guard timer would make it independent of the MCU.

6 Charge per event

Charge = current × time for each load, summed over the event. 1 μAh = 3.6 mC.

Event	What draws current	Charge (typ.)	μAh (typ.)	μAh (pess.)
Accepted card tap	Measured: field 12.2 ms, buzz 50 ms, Ready + LED to 250 ms, 1 empty poll	10.26 mC	2.85	3.01
Duplicate tap	Same read and poll, 2 × 60 ms motor at 82 mA + LED	15.19 mC	4.22	4.38
False wake-up	Measured: IRQ, 5.7 ms empty poll at 228 mA, re-arm	1.33 mC	0.37	0.42
Power-on (button)	Boot (20 ms at 80 MHz), reader start, 120 ms motor, 300 ms LED	11.05 mC	3.07	3.07
Lecture start (2-5 s hold)	3 s hold, 60 ms + 3 × 80 ms motor, 880 ms LED, marker write	28.32 mC	7.87	7.87
Auto power-off (3 min idle)	250 ms motor, 700 ms red LED, flush, reader power-down	22.63 mC	6.29	6.29
Power-off (5 s hold)	As above + 5 s hold + 60 ms hold buzz	29.61 mC	8.23	8.23
Short press (battery check)	0.2 s press, two 150 ms green blinks	0.91 mC	0.25	0.25
Flash page erase	22 ms at ~7 mA, once per 254 records	0.15 mC	0.04	0.04
Battery sample	1.5 ms MCU + ADC, every 10 s, on a heartbeat wake	1.8 μC	0.0005	0.0005

One accepted card tap, 10.3 mC (2.85 μAh)



Built from the scope captures. The buzz and the RF field (read plus one empty poll) are 82 % of a tap; the reader in Ready mode with the LED on for the rest of the pattern is most of the remainder.

Wake-up IRQ	= measured, TEK000006	= 0.026 mC
Read	= 202 mA × 12.2 ms (guard, REQA, anticollision, SELECT)	= 2.47 mC
Buzz	= 91 mA × 50 ms (motor, Ready, LED)	= 4.60 mC
Ready + LED	= 9.24 mA × 200 ms (rest of the pattern)	= 1.85 mC
Empty poll	= 228 mA × 5.7 ms + re-arm 14 μC	= 1.32 mC
Total	= incl. flash	= 10.26 mC = 2.85 μAh

7 A lecture, and battery life

The model lecture: the lecturer powers the unit on and starts a lecture with a button hold; 100 students tap over 15 minutes, 5 of them twice; 3 minutes after the last tap the unit switches itself off. It is on for 18 minutes.

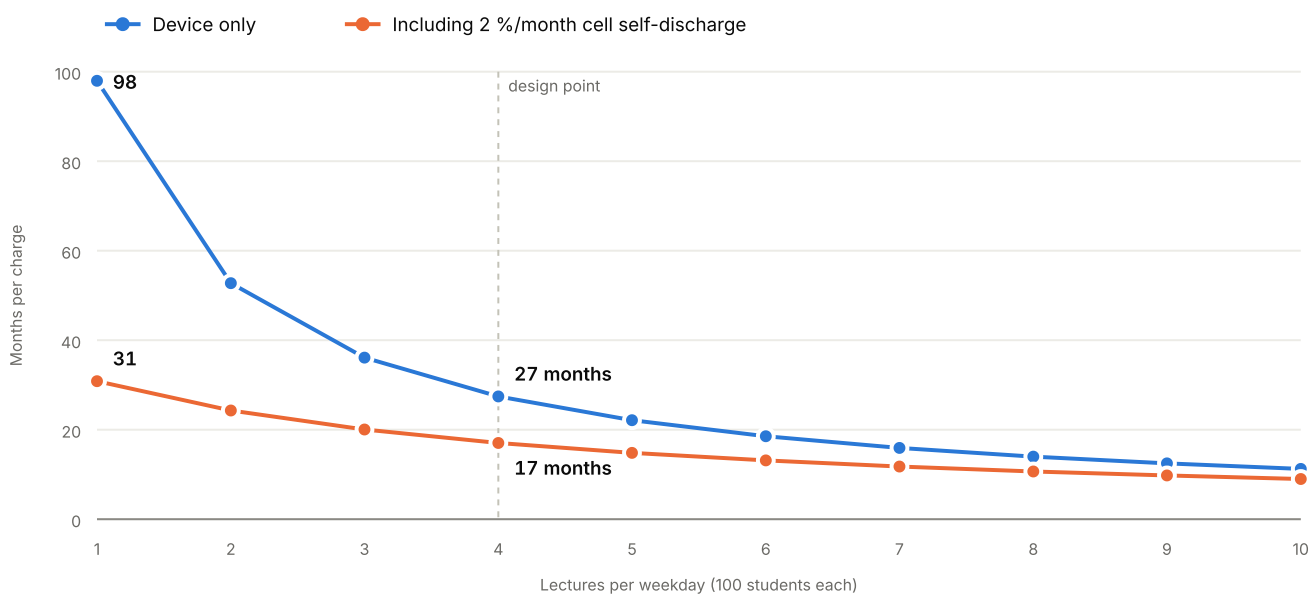
Part	μAh	Share
100 accepted taps	285.0	79 %
5 duplicate taps	21.1	6 %
False wake-ups (1/min)	6.6	2 %
Idle scanning, 18 min	32.5	9 %
Lecture start (button hold)	7.9	2 %
Auto power-off pattern	6.3	2 %
Power-on	3.1	1 %
Total per lecture	362	100 %

Per month at 4 lectures/weekday:

$$\begin{aligned}
 \text{lectures} &= 4 \times 21.7 &= 86.9 \\
 \text{device} &= 86.9 \times 0.362 \text{ mAh} \\
 &+ 1.80 \mu\text{A} \times 730 \text{ h} &= 32.8 \text{ mAh} \\
 \text{self-dis.} &= 2 \% \times 1000 \text{ mAh} &= 20.0 \text{ mAh} \\
 \text{total} &&= 52.8 \text{ mAh}
 \end{aligned}$$

$$\begin{aligned}
 \text{Life} &= 900 \text{ mAh} / 52.8 \text{ mAh} = 17.0 \text{ months} \\
 &\quad (\text{device only: } 27.4 \text{ months})
 \end{aligned}$$

Months per charge against lectures per day



Light use is limited by the cell's self-discharge, not the device. At 4 lectures a day a charge lasts about 17 months, with idle scanning and false wake-ups a large share of each lecture.

Scenario (weekdays)	μAh per lecture	Months, device only	Months, with self-discharge	Months, pessimistic
Light: 1/day × 50 students	207	155	35	23
Typical: 4/day × 100 students	362	27	17	12
Busy: 6/day × 150 students	518	13	10	8
Heavy: 8/day × 200 students	669	8	7	5

Shelf time when never switched on: $900 \text{ mAh} / 1.8 \mu\text{A} = 57 \text{ years}$ from the device's draw alone, but self-discharge empties the cell in roughly 3.8 years. Store the device charged and recharge it each term. All values are at room temperature; Li-ion capacity falls 10-20 % near 0 °C and with ageing, partly covered by the usable-capacity factor.

8 Card reads per charge

A tap alone costs 2.85 μAh , so a charge could fund **315,821** taps if nothing else drew current. With each lecture's idle time, power-on and power-off, duplicates and false wake-ups shared out among its students, the effective cost per student is higher, especially in small classes:

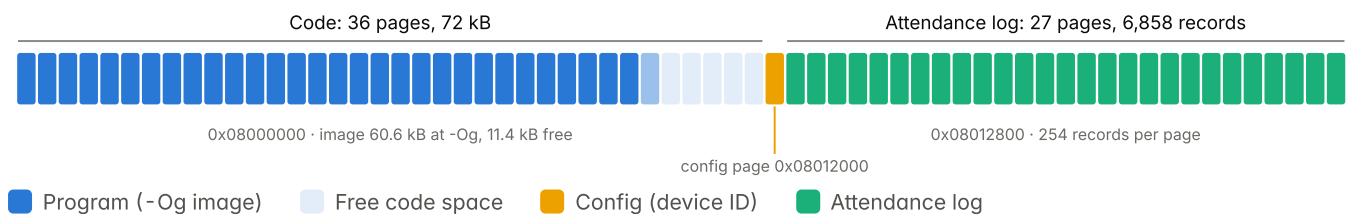
Usage pattern	μAh per student (all overheads)	Taps per charge (900 mAh)	Flash fills per charge
30 students per lecture	5.01	179,593	26x
60 students per lecture	4.00	224,952	33x
100 students per lecture	3.62	248,289	36x
200 students per lecture	3.34	269,238	39x

Even small classes get over 180 000 reads per charge (self-discharge aside), which is 26 to 39 times the 6,858 records the flash can hold. Before the battery is a limit, the log has to be read and cleared by the PC app many times over (by #CLEARLOG when it starts a lecture).

9 Flash: program and attendance log

The fitted MCU is the **STM32L432KB**: its flash-size word at 0x1FFF75E0 reads 128 (measured on the board). Its 128 kB is 64 pages of 2 kB, erased a page at a time, and the program and the log must share them. Earlier firmware assumed the 256 kB STM32L432KC and kept its data at 0x08020000, which the KB does not specify; the layout below fits the KB.

Flash map, one block per 2 kB page

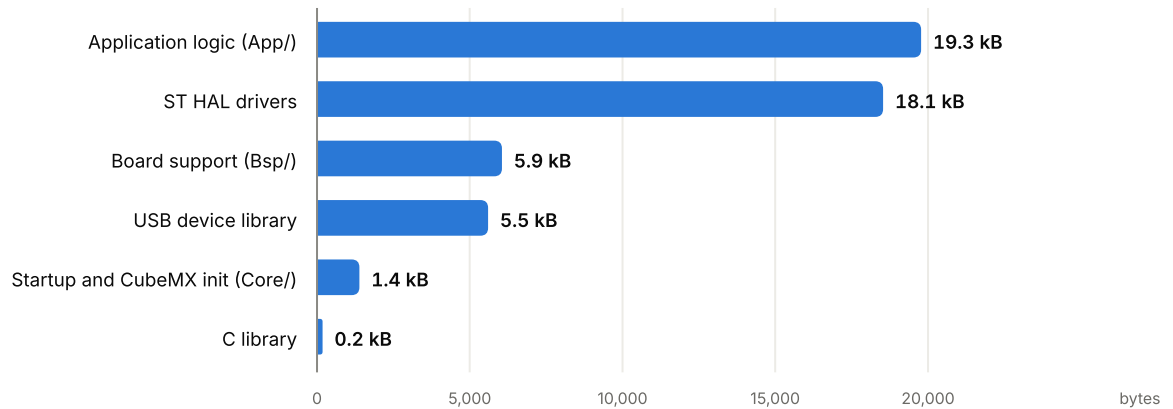


Program size

Measured with the project's own toolchain (arm-none-eabi-gcc 13.3, nano libc, unused sections removed):

Build	Image (bytes)	In 72 kB	Fits	Note
CubeIDE Debug before this change (-O0)	99,244	96.9 kB	No	Larger than any split that leaves a useful log
make and CubeIDE Debug (-Og)	62,052	11.4 kB free	Yes	84 % of the code region; full debugging
CubeIDE Release (-Os)	51,496	21.7 kB free	Yes	Shipping build
-Os with link-time optimisation	46,220	26.9 kB free	Yes	Not used: hardest to debug

Where the 50.3 kB -Os image goes



Application code and ST's HAL are about three quarters of the image. The HAL is used through its generic drivers; replacing it with direct register code could save several kB more, at the cost of portability.

Choosing the split

Every page given to code is 254 records fewer in the log. The CubeIDE Debug build used to compile at -O0 (97 kB), which fits no useful split, so it now builds at -Og like the Makefile. With -Og as the largest build that must fit:

Code / data	Log pages	Records	Lectures of 100	Headroom at -Og	Headroom at -Os
64 kB / 64 kB	31	7,874	74	3.4 kB	13.7 kB
72 kB / 56 kB (chosen)	27	6,858	64	11.4 kB	21.7 kB
80 kB / 48 kB	23	5,842	55	19.4 kB	29.7 kB

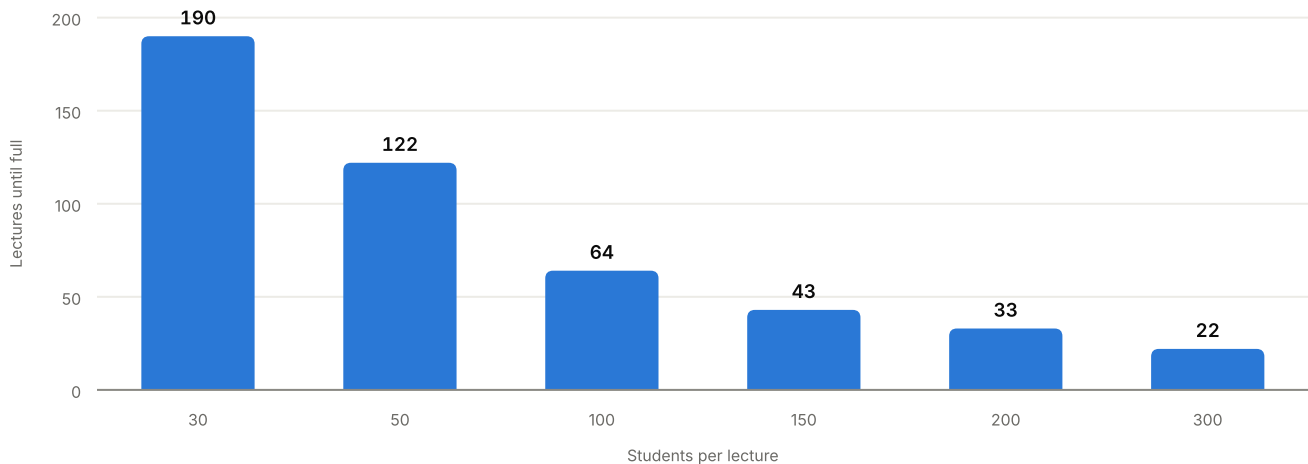
72 kB of code leaves 11.4 kB (16 %) for future features at -Og and still gives the log 6,858 records. A 64 kB code region would add 1,016 records (+15 %) but leave only 3.4 kB at -Og, forcing -Os for debugging. If records ever matter more than debugging, -Os with link-time optimisation (45.1 kB) would make the 64 kB split comfortable.

The linker script (STM32L432KBUX_FLASH.ld) gives the program 72 kB, so an image that outgrows it fails to link instead of overwriting the log. Link-time and compile-time checks keep the linker script, the board header and the log layout in step.

Log capacity

Page	= 2048 bytes = 256 double-words	
Records	= 256 - 1 header - 1 footer	= 254 per page
Log	= 27 pages × 254	= 6,858 records (8 bytes each: card ID + time)
Lecture	= 1 header + ceil((module + lecture + 2 NULs) / 4) text records	
	= 1 + ceil((6 + 10 + 2) / 4) = 6 for "EE4020" / "Lecture 12" (range 4-16)	
Lectures	= floor(6,858 / (N students + 6))	

Lectures the log can hold before a PC sync



Duplicates are not stored, so N counts distinct students in each lecture.

Students / lecture	Records / lecture	Lectures until full	Student taps stored	At 4 lectures per weekday
30	36	190	5,700	48 days (9.5 weeks)
50	56	122	6,100	30 days (6.1 weeks)
100	106	64	6,400	16 days (3.2 weeks)
150	156	43	6,450	11 days (2.1 weeks)
200	206	33	6,600	8 days (1.6 weeks)
300	306	22	6,600	6 days (1.1 weeks)

- **Students per lecture:** up to **6,852** in a single lecture after a clear (log size minus one marker).
- **Registered students:** not limited by the device. It keeps no card list; any 32-bit card ID below 0xFFFFFFFF00 is stored, and the PC app decides who each card belongs to.
- **When the log is full** new taps are refused with the red error pattern, because APP_LOG_WRAP_WHEN_FULL is 0. Up to 128 records wait in RAM (flushed at 80 % or 5 s after the last tap) and are written before power-off.
- **USB view:** a full log is 6,858 rows × 32 bytes = 219 kB of ATTEND.CSV (429 clusters), well within FAT12's 4084-cluster limit.
- **Flash wear:** each log page is erased once per fill-and-clear cycle. At the STM32L4's 10,000-cycle endurance, that is 10,000 full logs (about 69 million records): not a limit in practice.
- **Moving a unit to this layout:** the new firmware does not read the old data at 0x08020000. Import the unit's log with the PC app before reflashing, and set its device ID again afterwards.

10 Where to save more power

Ranked by effect. The reader's wake-up mode dominates idle current, so its period and the antenna tuning matter most; the rest are small.

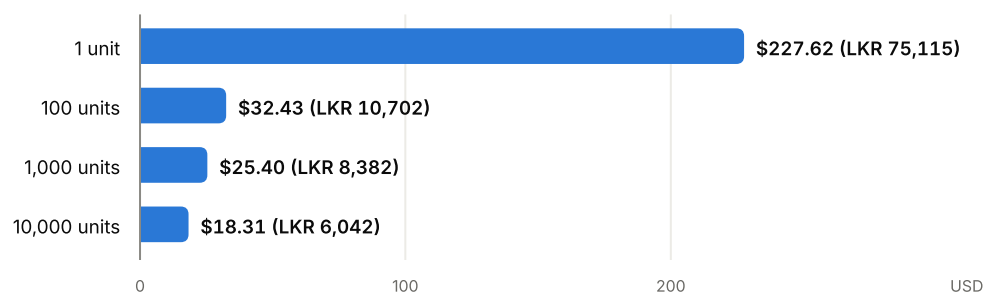
Change	Saving	Effect
Wake-up period 100 ms to 200 ms	- 50 μ A idle	46 % of idle; a card is noticed up to 100 ms later
Wake-up period 100 ms to 400 ms	- 74 μ A idle	69 % of idle; up to 400 ms to notice a card
Tune the antenna: field 203 mA to ~100 mA	- 1.8 mC per tap	18 % of each tap; probably cheaper wake-up measurements too
SPI prescaler /8 to /2	- 0.28 mC per read	3 % of each tap
Motor pulse 60 ms to 40 ms	- 1.6 mC per tap	16 % of each tap
Heartbeat off	- 5.5 μ A idle	5 % of idle; no sign the unit is on

11 Manufacturing cost

What a finished unit costs to make and land in Sri Lanka: the assembled board, the 1000 mAh cell, the motor, the case and box, freight from China, import charges, and final assembly and test in Sri Lanka. Prices are in US dollars with rupees at 1 USD = 330 LKR. The model is `Firmware/docs/cost_model.py`; it reads the LCSC BOM workbook in `PCB/bom`.

USD unless stated	1 unit	100 units	1 000 units	10 000 units
Electronic parts (LCSC)	53.97	1,283	11,197	98,380
PCB fabrication	2.00	30.00	240.00	1,800
PCB assembly (JLCPCB)	76.13	146.61	716.37	5,228
Battery, motor, case, box	16.00	895.00	6,700	34,500
Freight to Colombo	30.00	60.00	350.00	1,500
Import duty, VAT, levies (25 %)	44.52	603.69	4,801	35,352
Final assembly and test (Sri Lanka)	5.00	62.50	633.33	2,683
Yield loss	-	162.16	761.97	3,662
Order total, USD	227.62	3,243	25,399	183,106
Order total, LKR	75,115	1,070,226	8,381,667	60,424,952
Cost per unit, USD	227.62	32.43	25.40	18.31
Cost per unit, LKR	75,115	10,702	8,382	6,042
Price per unit at 40 % margin, USD	379.37	54.05	42.33	30.52
Price per unit at 40 % margin, LKR	125,192	17,837	13,969	10,071

Cost per unit against order size



A single unit carries a whole minimum order: 5 PCBs, 2 assembled boards, JLCPCB's setup and 20 Extended-part fees (\$61), and a courier. From 100 units the parts dominate, led by the STM32 and the ST25R3916.

Where the money goes

- **1 unit:** assembly fees (\$76 for the order) and parts in minimum lots (\$27 a board). Hand-soldering the prototype saves the assembly fees.
- **100 units:** parts are \$12.83 a board; the two ICs are over half.
- **1 000 units:** the case tooling adds \$3.00 a unit; parts fall to \$11.20.
- **10 000 units:** parts \$9.84; import charges are the second largest line.
- **Selling price** = cost / (1 - margin); the table shows a 40 % gross margin as an example.

Key part prices

Part (USD each)	1	100	1 000	10 000
STM32L432KBU6 (U1)	5.075	3.530	3.247	2.950*
ST25R3916-AQWT (U5)	5.440	3.753	3.400*	3.000*
MCP73833 charger (U2)	1.902	1.240	1.118	1.000*
TPS7A0233 LDO (U3)	0.557	0.360	0.335	0.300*
USB4110 USB-C (J1)	1.396	0.824	0.720*	0.620*
PTS636 switch, each (SW1-3)	0.356	0.257	0.187	0.177
NTC 10k (TH1)	0.626	0.385	0.311	0.280*
All electronic parts, per board	26.98	12.83	11.20	9.84

* extrapolated (LCSC lists no break at that quantity).

Assumptions

Item	1 / 100 / 1 000 / 10 000	Basis
Exchange rate	1 USD = 330 LKR	Rate on 3 October 2026; it moved 328-332 over the month before
Electronic parts	LCSC, 8 October 2026	Price breaks above; other parts at the BOM's minimum-lot price × 1, 0.85, 0.65, 0.5
PCB	\$2 for 5; then \$0.30 / \$0.24 / \$0.18	2 layers, 60 × 89 mm, per board (estimate)
Assembly	JLCPCB Economic to 100, Standard above	265 SMT + 18 hand-soldered joints, 20 Extended part types at \$3.07 each per order
Battery	\$6.00 / \$3.00 / \$2.20 / \$1.80	1000 mAh LiPo with protection board and JST XA lead (estimate)
Motor	\$1.00 / \$0.45 / \$0.30 / \$0.25	10 mm coin ERM with leads (estimate)
Case	\$8.00 / \$5.00 / \$3.90 / \$1.20	3D printed to 100; aluminium tool (\$3 000) at 1 000; steel tool (\$6 000) at 10 000
Freight	\$30 / \$60 / \$350 / \$1,500	Whole order, China to Colombo; lithium cells ship as dangerous goods
Import charges	25 % of CIF	Duty (HS code dependent) + 18 % VAT + 2.5 % SSCL + levies; VAT is reclaimable when registered
Labour	\$2.50/h; 120 min / 15 min / 8 min / 5 min	Fit battery, motor and case, flash, test; plus a \$300-600 test jig from 1 000

Item	1 / 100 / 1 000 / 10 000	Basis
Yield	100 % / 95 % / 97 % / 98 %	Units that pass test

Risks and exclusions. LCSC held only 27 ST25R3916 on 8 October 2026 and Digi-Key none, so 100 units or more need a quote from ST or a distributor. The duty rate depends on the HS code the goods are cleared under. Not included: the optional PN532 module, a USB cable, radio type approval (TRCSL, for a 13.56 MHz transmitter), firmware and engineering time, warranty and distribution. R6 is still 1 k Ω (1 A charge current); the BOM's 2 k Ω alternate (500 mA) suits a USB-C sink and costs the same.

Sources: lcsc.com product pages C2928224, C2908147, C627668, C2887324, C5143397, C2689642, C141588 (8 October 2026); jlcpcb.com/help/article/pcb-assembly-price; jlcpcb.com/news/discounts-for-large-pcb-orders; exchange-rates.org (USD/LKR); customs.gov.lk "Computation of import levies" (2026); KPMG, Sri Lanka budget 2026 tax proposals; teamrapidtooling.com (mould costs); besomi.com (603450 LiPo retail price).

12 Verification plan

- Fit and tune the antenna (C16-C19, R15), then repeat modes 2 and 4 of the power-test build: the field current and the wake-up mode average should both fall.
- Read dbg_nfc_false_wakes after an hour of idle with no card near (plug in USB to keep the MCU awake for the probe) and replace the assumed 1 per minute.
- Capture the end of a tap (the rest of the LED pattern and the empty poll, 100-300 ms after the card) on the scope; the model takes it from the false wake-up and the firmware's pattern length.
- Settle the wake-up mode average (99 μ A from the scope, 185 μ A from the multimeter) with a current-integrating meter (Nordic PPK2 or similar) over a minute of idle, with no debugger attached.